

Notes on The Effect: An Introduction to Research Design and Causality by Nick Huntington-Klein

Chapter 1 - Designing Research

1.1 I Have a Question

What is a research question

A research question is a clear, focused question that your study is designed to answer. It states exactly what issue, relationship, or phenomenon you want to investigate.

Key elements:

- It is based on a research problem or gap in existing knowledge and points to what needs further investigation.
- It guides the entire research process: what data you collect, which methods you use, and how you analyze and interpret results.

Good research questions are:

- Specific, concise, and understandable to others without extra explanation.
- Researchable (you can answer them with data) and open-ended enough that the answer is not trivial or already known.

Research question types

There are several common ways to classify research question types. The main ones you'll see in methodology texts are:

- **By logical aim:**

- **Descriptive** – ask “what is...?” or “how much/how many...?” and aim to describe characteristics, patterns, or prevalence.
 - * Example: “What are the main factors influencing customer churn in telecom companies?”
- **Relational / correlational** – ask about relationships between variables, without necessarily claiming causality.
 - * Example: “Is there a relationship between weekly screen time and self-reported stress levels among adults?”
- **Causal / explanatory** – ask whether one variable causes or affects another.
 - * Example: “What is the effect of participation in a data-literacy training program on analysts’ report accuracy?”
- **By data paradigm**
 - **Quantitative research questions** – framed around variables, measurement, and statistical relationships (e.g. “To what extent...?”, “What is the effect of X on Y...?”).
 - **Qualitative research questions** – framed around meanings, experiences, processes (e.g. “How do...?”, “Why do...?”, “In what ways...?”).

Many guides first ask “Is your study quantitative, qualitative, or mixed-methods?” and then shape the question accordingly (e.g. quantitative descriptive vs. qualitative exploratory).

More fine-grained types you may see:

Different authors and courses introduce slightly richer taxonomies; common labels include:

- **Comparative** – focus on differences/similarities between groups, conditions, or time periods.
 - Example: “How does job satisfaction differ between remote and on-site data analysts?”
- **Predictive** – focus on forecasting or prediction accuracy.
 - Example: “To what extent can historical purchase behavior predict future churn among retail customers?”
- **Exploratory** – used in under-researched areas; broad, open-ended questions to identify patterns or themes.
 - Example: “How do junior data scientists describe the main challenges in adopting MLOps practices?”

- **Evaluation** – assess effectiveness, impact, or outcomes of a program, policy, or intervention.
 - Example: “How effective is a weekly code-review practice in reducing production bugs in analytics pipelines?”
- **Action-oriented / design / improvement** – aim at finding solutions or informing change.
 - Example: “What process changes can reduce report refresh time in a Power BI environment?”

1.2 Empirical Research

What is empirical

Empirical means “based on experience, observation, or experiment,” rather than only on theory.

- **Core meaning**
 - Empirical data: data you actually observe or measure in the real world (experiments, surveys, logs, etc.), not just results from assumptions or theory.
 - Empirical approach: making conclusions from evidence and experience, instead of purely from abstract reasoning.
- **In science and data work**
 - An empirical study is a study that collects real-world data (observations, experiments, questionnaires) and analyzes it to answer a question or test a hypothesis.
 - In contrast, a purely theoretical study works with models, proofs, or simulations that are not directly tied to observed data.

One particularly sticky problem with quantitative empirical research is that the numbers that we observe often don't tell us exactly what we want to know.

1.3 Why Research Needs a Design

Why is it so important for research to be properly designed? One way we can think about this is by looking at what happens when it's not.

? Research in an empirical context needs a design because the design is what turns an interesting idea or question into a coherent, valid, and reproducible study that can actually answer that question.

1.4 In This Book

Part I focuses on the logic of design: how to formulate a clear, answerable causal question and then reason about what kind of quantitative empirical study could address it. It guides you to decide what needs to be measured and how to check that your proposed approach really targets the question you care about.

Part II turns to the methods toolbox: it presents standard causal inference techniques for using observational data when you cannot run an experiment. These methods are widely used because they work in many applied settings and rest on assumptions that are now well studied and relatively well understood.

Chapter 2 - Research Questions

2.1 What Is a Research Question?

Coming up with a question is easy. Coming up with a good research question is much harder.

What's the difference? The difference, at least in the case of quantitative empirical research, is that a research question is a question **that can be answered**, and for which having that answer will **improve your understanding of how the world works**.

What does it mean to have a question that can be answered? It means that it's possible for there to be some set of evidence in the world that, if you found that evidence, your question would have a believable answer.

So we have a question that can be answered. But does it improve our understanding of how the world works? What this means is that the research question, once answered, should tell you about something broader than itself. It should inform theory in some way. Theory doesn't have to be something as important as the theory of gravity or the theory of evolution. It could even be something as generic as "bread costs more today than last year because bread prices have been generally increasing over time." Theory just means that there's a why or a because lurking around somewhere. Even hydrogen is a theory - it says that we see material like water behaving a certain way because there's a kind of atom that behaves in certain ways and has a certain structure.

Theory - An explanation of why the things we observe happen, or otherwise generalizes what we observe in one situation to another situation.

A good research question takes us from theory to hypothesis, where a hypothesis is a specific statement about what we will observe in the world, like "people who wash their hands will get sick less often."

Hypothesis - A specific statement about what you will observe in the data.

2.2 Why Start with a Question?

There's data everywhere. So why not skip the hard part of deriving a research question from a theory and instead just see what sorts of patterns are in the data?

This is called "data mining".

The kinds of things that data mining is good at are in finding patterns and in making predictions under stability. The kinds of things that data mining is less good at are in improving our understanding, or in other words helping improve theory. It also has a tendency to find false positives if you aren't careful.

What do I mean by "under stability"? I mean that the process giving us the data doesn't change. If I roll a six-sided die a thousand times, data mining would be great at predicting that the probability of a 1 is. But if I then switch to a twenty-sided die, that data mining prediction will be bad. It will still predict a chance of a 1 until it gets a lot more data. Probability theory, on the other hand, will properly predict the switch to a chance immediately.

Sometimes seeing patterns in data can give us ideas for research questions that we can examine further in other data sources.

It's also probably the best angle to take when we don't care about why! If I don't care why the stock market goes up or down, and I just want to predict if it will or not so that I know whether to buy or sell, then data mining may well be the way to go.

There are ways of avoiding false positives while doing data mining - this is something they worry about a lot in data science and have a lot of tools for.

Some examples are "cross-validation" and "training and testing sets."

That said, data mining isn't all bad. There's no way we could possibly think up every interesting theory to test. Plenty of theories come from looking at the data in the first place, noticing a pattern, and wondering why the pattern is showing up, or whether the pattern is even real.

Of course, the responsible thing to do at that point is to not just take the pattern as given. That's where the real problem of data mining is. Instead, take the interesting pattern noticed and look to see if it holds up in other data - if it replicated - before being sure that the pattern noticed is real and explains how what you saw worked (Viagra example).

2.3 Where Do Research Questions Come From?

Research questions can come from lots of places. Mostly, curiosity.

- theory → hypothesis → question
- data → opportunity → question

2.4 How Do You Know if You've Got a Good One?

- Consider Potential Results - If you can't say something interesting about your potential results, that probably means your research question and your theory aren't as closely linked as you think.
- Consider Feasibility - A research question should be a question that can be answered using the right data, if the right data is available. But is the right data available?
- Consider Scale - What kind of resources and time can you dedicate to answering the research question?
- Consider Design - A research question can be great on its own, but it can only be so interesting without an answer.
- Keep It Simple! - Answering any research question can be difficult. Don't make it even harder on yourself by biting off more than you can chew! A common mistake is to bundle a bunch of research questions into one.

So, consider feasibility, scale, and design. Keep it simple, and think about whether the results you might likely see would tell you anything interesting about the world. After all, learning something interesting and new about the world is our goal!

Homework for Chapter 2: Research Questions

1. What is data mining useful for? What are its limitations?

Core purposes

Data mining helps identify patterns, correlations, and anomalies that are not obvious through simple descriptive analysis, using machine learning, statistics, and AI. These patterns can then be used to understand current behavior (diagnostic/analytic) and to forecast future outcomes (predictive).

In research, data mining is very useful for finding correlations and associations, but it does not, by itself, establish causation; you need explicit causal designs or models on top of the mined patterns. Used carefully, it can generate hypotheses about possible causal mechanisms that you then test with appropriate methods (experiments, quasi-experiments, causal inference frameworks).

Correlation, association, and data mining

Data mining excels at uncovering **associations** and **correlations** in large, complex datasets by systematically scanning for regularities, co-occurrences, and predictive relationships. This includes tasks like association rule mining, clustering, and predictive modeling, which reveal “X tends to occur with Y” or “Y can be predicted from X” without assuming any causal direction.

A key issue is that data mining can easily produce large numbers of **spurious correlations** — patterns that are statistically real but substantively meaningless. **Correlation does not imply causation** is especially critical in the context of big data and automated pattern search, because sheer volume guarantees you will find many misleading associations.

Why correlation causation

Correlation (or association) means two variables tend to vary together, while causation implies that changes in one variable directly bring about changes in another. In scientific terms, causation requires a mechanism plus a set of conditions (temporality, non-spuriousness, and, often, counterfactual reasoning) that can't be confirmed simply by observing co-variation in observational data.

Big data and data mining increase statistical power and make it easier to detect correlations, but they do not remove core identification problems such as confounding, selection bias, and reverse causality. There is ongoing debate about whether big data alone can “upgrade” correlations to causation, with most methodological work agreeing that you still need strong research designs or causal models, not just more data.

Role of data mining in research workflows

In research settings, data mining is typically used as:

- A hypothesis generation tool: scanning large datasets to find interesting, unexpected correlations that may suggest new causal questions (“Could X be influencing Y?”).
- A variable and feature discovery tool: identifying promising predictors, interactions, and latent groupings, which can then be formalized in statistical or causal models.
- A pattern characterization tool: providing rich descriptive insight into how phenomena cluster or co-vary across populations, contexts, or time.

Data mining and causal inference

There is growing work on combining data mining with causal inference, but the causal claims come from the identification strategy, not from the mining step itself. Approaches include:

- Using data mining to select covariates, identify strata, or discover heterogeneity, then applying causal methods such as matching, inverse probability weighting, difference-in-differences, or instrumental variables.
- Embedding causal graphs (DAGs) or potential-outcomes reasoning into machine-learning workflows, so that algorithms respect causal assumptions (e.g., learning treatment effect heterogeneity rather than just prediction accuracy).

Practical implications for research

For empirical work, a useful way to frame it is: data mining is excellent for **finding and summarizing correlations**, but you should treat those as inputs into a separate causal analysis pipeline. Typical best practices include:

- Explicitly distinguishing exploratory (data mining) from confirmatory (causal testing) phases in your project documentation and analysis plans.
- Using mined patterns to refine theory, operationalize constructs, and build candidate models, while reserving strong causal language (“effect”, “impact”, “causes”) for results backed by identification strategies.
- Guarding against spurious correlations by cross-validation, out-of-sample checks, and, where possible, triangulation with experiments, quasi-experiments, or longitudinal designs.

*A **quasi-experiment** is a study that tries to estimate causal effects by manipulating an intervention, but without random assignment of units to treatment and control groups. It “resembles” a true experiment in structure, yet lacks at least one key experimental safeguard—usually randomization.*

Core definition

In a quasi-experimental design, the researcher still defines an independent variable (the “treatment” or intervention) and observes its effect on a dependent variable, but participants or units end up in groups based on real-world criteria rather than random assignment. This is why quasi-experiments are often described as “natural” or “real-world” experiments: the researcher leverages existing groups or policies instead of creating them randomly.

How it differs from a true experiment

In a classic randomized controlled trial (RCT), randomization ensures that, on average, treatment and control groups are equivalent on both observed and unobserved characteristics at baseline. In a quasi-experiment, group differences can exist before the intervention (e.g., different schools, regions, firms, time periods), which introduces threats to internal validity. As a result, quasi-experimental causal claims rely on design and analytic adjustments—rather than randomization—to argue that alternative explanations (confounding, selection bias) have been minimized.

Typical quasi-experimental designs

Common quasi-experimental structures include:

- *Nonequivalent control group designs: compare outcomes between a “treated” group and a similar but non-random control group (e.g., two schools, one adopts a new curriculum, one does not).*

- *Interrupted time series designs: examine outcomes measured repeatedly over time, before and after an intervention or event, to see whether there is a clear “break” in the trend at the intervention point.*
- *Difference-in-differences and related panel designs: compare changes over time between treated and untreated groups, assuming that in the absence of treatment they would have followed parallel trends.*

All of these manipulate or at least exploit an intervention but use design logic (timing, comparison groups, trends) instead of random assignment to make causal inferences.

Why quasi-experiments are used

Quasi-experiments are especially valuable when randomization is impossible, unethical, or politically infeasible—for example:

- *Evaluating a nationwide policy that applies to everyone in a particular jurisdiction.*
- *Studying the impact of natural events (economic shocks, disasters, pandemics).*
- *Assessing organizational changes where management will not randomize units.*

They give you a structured way to approximate experimental logic in field conditions, which is why they are heavily used in public policy, education, labor markets, health services, and other applied social science settings.

Causal inference and limitations

From a causal-inference standpoint:

- ***Strength:*** *Quasi-experiments are usually much stronger than purely observational cross-sectional correlations because the intervention precedes the outcome and the design explicitly leverages comparison groups or time patterns.*
- ***Weakness:*** *They are more vulnerable than RCTs to confounding and selection bias, since units self-select or are administratively assigned to conditions. Researchers must therefore carefully justify assumptions (e.g., parallel trends, no unmeasured confounders that change at the intervention point) and often perform robustness and sensitivity analyses.*

Association rule mining *is a data mining technique for discovering “if-then” patterns of co-occurrence between items in large datasets, typically expressed as rules like $X \rightarrow Y$ (“if X occurs, Y tends to occur too”). It’s a classic unsupervised method used to uncover associations rather than to predict a labeled outcome.*

Core idea

Association rule mining scans collections of transactions or events (e.g., shopping baskets, page visits, diagnoses) to find item combinations that occur together more often than you’d expect

by chance. Each discovered pattern is formalized as an association rule with an antecedent (left-hand side, the “if” part) and a consequent (right-hand side, the “then” part).

Example in market-basket data:

- Rule: $\{\text{bread, butter}\} \rightarrow \{\text{jam}\}$ This can be interpreted as: in transactions where bread and butter are bought, jam is frequently bought as well.

Key measures: support, confidence, lift Three core metrics are used to evaluate rules:

- Support: How frequently a set of items appears in the dataset (joint occurrence).
- Confidence: The conditional probability of the consequent given the antecedent; intuitively, how often the rule is correct when the antecedent holds.
- Lift (and related measures): How much more often antecedent and consequent occur together than you would expect if they were independent; used to filter out rules that are high-confidence but trivial or spurious.

Frequent itemsets are first identified based on minimum support, and then association rules are generated from those itemsets that also meet minimum confidence (and often lift) thresholds.

Algorithms commonly used

Classic algorithms for association rule mining include:

- Apriori: Level-wise search over itemsets, using the property that all subsets of a frequent itemset must themselves be frequent to prune the search space.
- FP-Growth: Builds a compact tree structure (FP-tree) of frequent patterns and mines itemsets without generating as many candidate sets as Apriori.
- Eclat: Uses a vertical data format (item $\rightarrow$ list of transaction IDs) and intersection of transaction ID sets to find frequent itemsets.

In practice, these algorithms are wrapped in libraries (e.g., the *arules* package in R) and tuned via *minsup*, *minconf*, and sometimes *minlift* to control the number and quality of rules.

Typical use cases

Association rule mining is widely used for:

- Market basket analysis: Product bundling, shelf placement, cross-selling (“customers who buy X often buy Y”).
- Recommender systems: Suggest related items or content based on co-view/co-purchase patterns.
- Web and app analytics: Common navigation paths, sequences that lead to conversion or drop-off.

- *Healthcare and operations: Co-occurring symptoms, tests, or procedures; common failure combinations in equipment logs.*

Relation to correlation and causation

Conceptually, association rules describe correlations/associations, not causation: they tell you that items co-occur, not that one causes the other. For research workflows, they are typically used for:

- *Exploratory analyses and hypothesis generation.*
- *Discovering interesting behavioral or operational patterns that can later be tested in more formal, causal designs.*

2. Are each of the following questions a good research question? Explain your reasoning. If the question is not good, suggest ways to make it a better question if possible.

- Do severe hurricanes cause psychological disorders in children in gestation during the hurricanes?

Is it a good research question?

Mostly yes: it's causal, focused on a specific exposure (severe hurricanes) and outcome (psychological disorders), and a defined target group (children in gestation during the hurricanes). The main issues are lack of operational definitions and feasibility.

Points in favor:

- Clear causal structure: exposure (hurricane during gestation) → outcome (later psychological disorders).
- Population is at least implicitly defined (children who were in utero at the time of the hurricane).

Limitations:

- "Severe hurricanes" is not operationalized (e.g., category, wind speed, economic damage, proximity of residence).
- "Psychological disorders" is broad; clinical diagnosis, screening scores, age at assessment, and specific disorders would need specification.

No time frame (e.g., assessed at age 5, 10, adolescence).

How to make it better:

Does prenatal exposure to Category 3 or higher hurricanes increase the incidence of clinically diagnosed anxiety and depressive disorders at age 10 among children whose mothers lived in affected coastal regions?

- Exposure defined: Category 3+ hurricanes, coastal residence.
- Outcome defined: incidence of diagnosed anxiety/depressive disorders.
- Time frame: assessment at a specific age.
- Population: children of mothers in affected regions.

For causal inference, you would then think about design (e.g., difference-in-differences comparing cohorts gestating before/after versus during hurricane, controlling for confounders like SES, maternal mental health).

SES means socioeconomic status is a composite indicator of a person's or family's social and economic position, typically captured with variables such as:

- *Income (household or individual)*
- *Education level*
- *Occupation or employment status*
- *Sometimes neighborhood characteristics or wealth*

In the above example, SES would be used as a confounder because families with different socioeconomic status may both:

- *Be differentially exposed to severe hurricanes (e.g., living in more vulnerable housing or areas), and*
- *Have different baseline risks for children's psychological outcomes (through access to health care, stress levels, nutrition, etc.).*

- Are dogs the best pets?

Is it a good research question?

No, as written this is not a good empirical research question. It's vague, value-laden, and lacks defined constructs. "Best" is a subjective judgment and "pets" is a large, heterogeneous category.

Problems:

- Ambiguous outcome: "best" could mean satisfaction, health benefits, ease of care, cost, emotional attachment, etc.
- No specified population (children, older adults, urban households, people with allergies, etc.).
- No comparison structure: which other pets (cats, fish, reptiles, "all others")?
- Strong normative tone rather than a neutral, testable question.

How to make it better:

One way is to convert it into a comparative, well-defined question with measurable outcomes:

- “Do adult pet owners who keep dogs report higher life satisfaction than those who keep cats?”
- “Are rates of daily physical activity higher among dog owners than among owners of other companion animals?”
- “Do households with dogs incur higher monthly pet-related expenses than households with cats?”

These revised questions:

- Define outcome: life satisfaction, physical activity, expenses.
- Specify population: adult pet owners, households.
- Describe comparison: dogs vs. cats vs. other pets.

It could also be framed more explicitly as multi-criteria evaluation:

“How do dog, cat, and small-animal ownership compare in terms of owner satisfaction, time investment, and annual cost among urban adults?”

That keeps the comparative flavor without asserting “best” up front and supports multi-dimensional analysis.

- Do organic labels in grocery store items increase sales of those items?

Is it a good research question?

Yes, this is close to a good causal research question. It’s concise, clearly causal, and focuses on a measurable outcome (sales) and a specific intervention (organic labels).

Strengths:

- Clear causal framing: organic labels → change in sales volume or revenue.
- Variables are measurable: presence of organic label, sales data.
- Direct relevance to marketing, consumer behavior, and retail practice.

Limitations:

- “Organic labels” is not fully specified (certification body, design, placement, whether price changes accompany the label).
- “Sales” needs clarification: units sold, revenue, market share, basket penetration, time window.

- Population/context is missing: specific product categories, stores, regions, or online vs. brick-and-mortar.

How to make it better:

Tighten the scope and measurement:

“Does adding certified organic labels to private-label packaged food products in supermarket chains increase unit sales over the subsequent six months compared to similar non-labeled products?”

Or, for an experimental design:

“In a controlled online grocery experiment, does displaying an organic certification logo on a product image increase the probability of purchase compared to the same product without the logo?”

These versions:

- Specify context: supermarket chains or online grocery.
- Define exposure: adding certified organic labels/logo.
- Define outcome: unit sales or purchase probability in a defined period.
- Allow clear identification and appropriate causal design (e.g., A/B test, difference-in-differences using label introduction timing).

For defining a good research question, a useful pattern would be:

“Does [specific exposure or intervention] lead to [specific, measurable outcome] in [defined population] over [defined time frame]?”

- **fits this pattern with more operational precision.**
- **needs transformation from “best” into specific comparative outcomes.**
- **already follows the pattern and mainly needs tighter definitions and context.**

3. Think about a possible research question in your field of study (e.g., your major).

- a. Can this question be answered? Is it feasible to answer it?
- b. Does the question answer something about how the world works? If so, how?
- c. Write down some hypotheses based on the question.
- d. What kind of data needs to be collected to test the hypothesis?

4. For each of the following theories and research questions, does the research question inform theory (that is, does answering the research question tell you something about the theory?).

Theory:

- People donate to political campaigns largely because they want to defeat a political opponent, not because they like the candidate they're donating to.

Research question:

- Do negative advertisements increase political campaign donations in US Congressional races more than positive advertisements do?

Does the question inform the theory:

Yes, it's quite well aligned. If negative ads (which focus on attacking an opponent) increase donations more than positive ads (which emphasize liking or supporting the candidate), that pattern supports the idea that donations are primarily motivated by a desire to defeat the opponent rather than by admiration for "our" candidate.

If you find that negative ads substantially boost donations relative to positive ads, that is evidence consistent with the theory.

If positive ads are equally or more effective, that would challenge the theory (suggesting liking/support may matter as much or more than opposition).

So, answering this question would tell you something meaningful about whether the motivational emphasis in donations is "anti-opponent" versus "pro-candidate."

Theory:

- Men have higher average salaries than women in the US partially because employers discriminate against women in job interviews

Research question:

- Does the use of a gender-blind interview process, where hiring decisions and salary offers are made without knowing the employee's gender, shrink the salary gap between men and women?

Does the question inform the theory?

Yes, very directly. The theory posits discrimination in interviews as a causal mechanism contributing to the gender salary gap. A gender-blind interview is, in effect, an intervention that removes the proposed mechanism at hiring.

If introducing gender-blind interviews leads to a smaller salary gap (controlling for other factors), that supports the idea that discrimination in interviews is one reason for the gap.

If the gap remains unchanged, that suggests either (a) discrimination is happening at other stages (promotion, performance evaluation, negotiation) or (b) other mechanisms (e.g., occupational sorting, hours worked) are more important than interview discrimination.

So this question is a strong theory-informing test because it targets the specific causal channel the theory highlights.

Theory:

- Living in a rich country improves happiness.

Research question:

- Do people who live in richer countries on average report higher happiness levels?

Does the question inform the theory?

Yes, but somewhat coarsely. The theory is about causation (living in a rich country → higher happiness), while the research question is about cross-national association (richer countries higher average reported happiness).

If people in richer countries do report higher happiness on average, that pattern is consistent with the theory, but doesn't by itself prove causation (rich countries differ in many things besides income: institutions, culture, welfare systems, etc.).

If there is no association, or if poorer countries are as happy or happier, that would challenge the theory's claim that richness per se improves happiness.

So the question definitely informs the theory (it checks a key empirical implication), but it doesn't fully isolate the causal effect of wealth from other country-level factors. A more causally focused design (e.g., looking at changes over time as countries become richer, or using quasi-experimental shocks) would inform the theory even more strongly.

Theory:

- Many plants survive without eating because they collect energy from sunlight.

Research question:

- Are plants that live underneath a thick canopy (where little light can get through) less healthy than those in a loose canopy?

Does the question inform the theory?

Yes, indirectly but meaningfully. The theory says plants are sustained by energy from sunlight, so reduced light should, in general, reduce plant health (all else equal).

If plants under thick canopy (i.e., with low light) are less healthy than those under loose canopy (more light), that is consistent with the theory that sunlight is a key energy source.

If they are equally healthy, or healthier, that would suggest either (a) these plants have strong adaptations (e.g., shade tolerance, alternative energy pathways), or (b) the simple statement “many plants survive because of sunlight” needs to be qualified.

This question doesn’t test the mechanism as directly as, say, controlled experiments varying light intensity, but the observed health differences under different light conditions still provide evidence relevant to the theory.

Chapter 3 - Describing Variables

3.1 Descriptions of Variables

Empirical research questions really come down entirely to describing the density distributions of statistical variables.

In order to make any sense of data we have to know how to take some observations and describe them. The way we do that is by describing the types of variables we have and the distributions they take. Part of that description will be in the form of describing how different variables interact with each other.

A variable, in the context of empirical research, is a bunch of observations of the same measurement - the monthly incomes of 433 South Africans, the number of business mergers in France in each year from 1984 to 2014, the psychological “neuroticism” score from interviews with 744 children, the color of 532 flowers, the top headline from 2,348 consecutive days of The Washington Post. Successfully describing a variable means being able to take those observations and clearly explain what was observed without making someone look through all 744 neuroticism scores themselves.

3.2 Types of Variables

Most common kinds of variables:

- **Continuous variables** are variables that could take any value (perhaps within some range).
- **Count variables** are those that count something. Perhaps how many times something happened or how many of something there are. The number of business mergers in France in a given year is an example of a count variable. Count variables can’t be negative, and they certainly can’t take fractional values. They can be a little tougher to deal with than continuous variables. Sometimes, if a count variable takes many different values, then it acts a lot like a continuous variable, and so researchers often treat them as continuous.

- **Ordinal variables** are variables where some values are “more” and others are “less,” but there’s not necessarily a rule as to how much more “more” is.
- **Categorical variables** are variables recording which category an observation is in. Categorical variables are very common in social science research, where lots of things we’re interested in, like religious affiliation, race, or geographic location, are better described as categories than as numbers.
- A special version of categorical variables is **binary variables**, which are categorical variables that only take two values. Often, these values are “yes” and “no.” Binary variables are handy because they’re a little easier to deal with than categorical variables, since they’re useful in asking about the effects of treatments (Did you get the treatment? Yes or no), and also because categorical variables can be turned into a series of binary variables.

Instead of our religious affiliation variable being categorical with options for “Christian,” “Jewish,” “Muslim,” etc., we could have a bunch of binary variables - “Are you Christian?” Yes or no. “Are you Jewish?”

Plus, it allows for things like someone in your data being both Christian and Jewish.

- **Qualitative variables** are a sort of catch-all category for everything else. They aren’t numeric in nature, but also they’re not categorical.

These can be very tricky to work with and describe, as these kinds of variables tend to contain a lot of detail that resists boiling-down-and-summarizing. Often, in order to summarize these variables, they first get turned into one of the other variable types above.

3.3 The Distribution

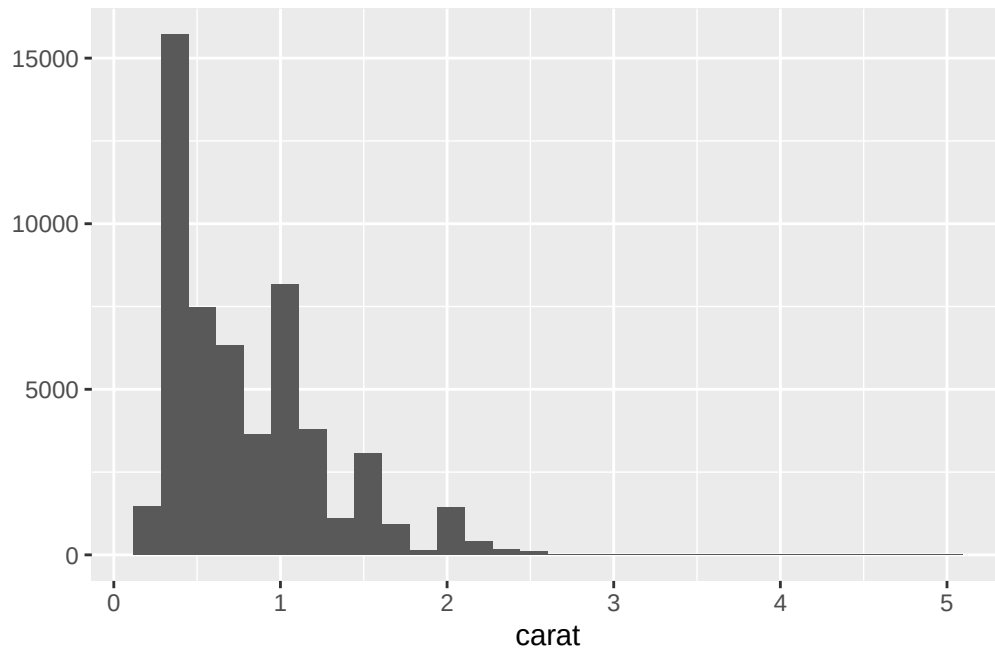
A variable’s distribution is a description of how often different values occur.

When it comes to **categorical or ordinal variables**, the variable’s distribution can be described by simply giving the percentage of observations that are in each category or value. The full distribution can be shown in a **frequency table or bar graph**, which just shows the percentage of the sample or population that has each value.

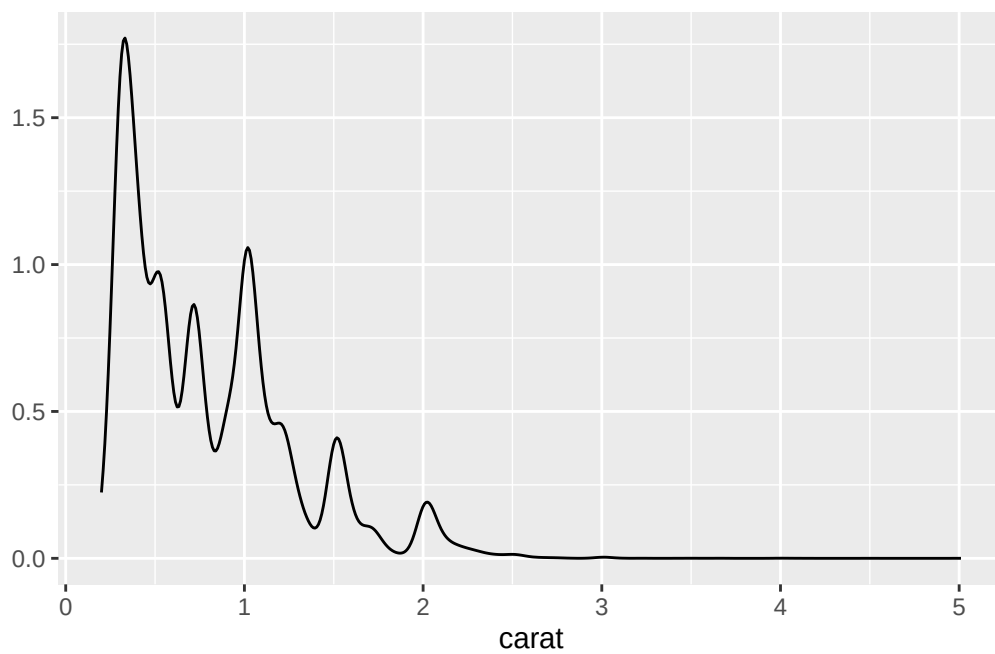
Continuous variables are a little trickier. We can’t just do a frequency table for continuous variables since it’s unlikely that more than one observation takes any specific value.

For **continuous variables**, distributions are described not by the probability that the variable takes a given value, but by the probability that the variable takes a value close to that one.

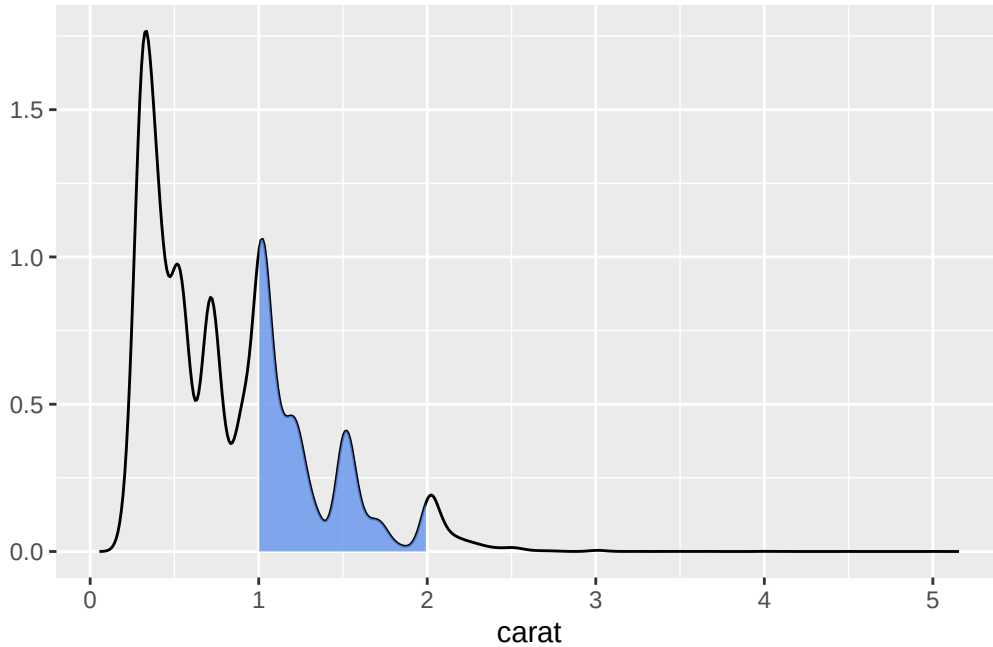
One common way of expressing the distribution of a continuous variable is with a **histogram**. A histogram carves up the potential range of the data into bins, and shows the proportion of observations that fall into each bin.



With a continuous variable, we can go one step further than a histogram all the way to a **density plot** (density distribution if you're graphing it, probability density if you're describing it).



A density shows what would happen to a histogram if the bins got narrower and narrower (this only works because we have enough observations to fill in each bin. Otherwise it looks a lot choppier. Formally we need the bins to get narrower and the number of observations to get bigger).



When we have a density plot, we can describe the probability of being in a given range of the variable by seeing how large the area underneath the distribution is.

That area, relative to the size of the area under the entire distribution curve, is the probability of the diamond being between 1 and 2 carats.

That particular shaded area makes up about 28% of the area underneath the curve, and so 28% of all diamonds have carats between 1 and 2. So, for each possible value the variable could take, we know how likely that outcome, or at least an outcome like it, is.

3.4 Summarizing the Distribution

A distribution is typically summarized by describing its center, spread, and shape, plus any outliers or special regions (like the 1–2 carat band).

Center (location)

Common measures:

- **Mean:** arithmetic average; good when the distribution is roughly symmetric and not too heavy-tailed.
- **Median:** value with 50% below and 50% above; more robust to skew and outliers.
- **Mode:** most frequent value or peak of a density/histogram; useful for multimodal data.

Together, these tell what a “typical” value looks like and whether the distribution is symmetric or skewed (e.g., mean > median suggests right skew).

Spread (variability)

Key summaries:

- **Standard deviation or variance:** average squared deviation from the mean; standard for symmetric, roughly normal distributions.

The way that variation shows up in a distribution graph is in how wide the distribution is.

If the distribution is tall and skinny, then all of the observations are scrunched in very close to the mean. Low variation. If it’s flat and wide, then there are a lot of observations in those fat “tails” on the left and right that are far away from the mean. High variation! The distributions with more area “piled in the middle” have little variation - not a lot of area far from that middle point! The distributions with less “piled in the middle” and more in the “tails” on either side have plenty of observations far away from the middle.

Variance measures average squared deviation from the mean. Standard deviation is the square root of variance.

Variance is convenient in some theoretical and modeling contexts (e.g., ANOVA, regression, some optimization problems) because squared deviations behave nicely algebraically.

Standard deviation is more commonly used for describing data because it’s on the same scale as the variable, so you can say things like “most observations are within about 10 units of the mean.”

In practice, the variance is often computed as an intermediate step, then standard deviation reported as the more interpretable summary.

- **Interquartile range (IQR):** 75th percentile minus 25th percentile; robust to outliers and skew.
- **Range or trimmed range:** min–max or central interval (e.g., 2.5th–97.5th percentiles).
- The other two percentiles to focus on are the **minimum, or the 0th percentile, and the maximum, or the 100th percentile**. These are the lowest and highest values the variable takes. They’re handy because they show you the kinds of values that the variable produces. The minimum and maximum height of a large group of people would tell you something about how tall or short humans can possibly be, for example.

- Another nice thing about the minimum and maximum is that we can take the difference between them to get the **range**. The range is one way of seeing how much a variable varies.

Shape and tails

Qualitative but important:

- **Skewness:** direction and strength of asymmetry (left/right skew).
- **Modality:** unimodal, bimodal, multimodal.
- **Tail behavior:** presence of heavy tails or extreme values.

Often this is described verbally (“right-skewed with one dominant mode around 1 carat”) alongside a histogram or density plot.